

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME : MLA03

COURSE CODE : Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

(5 Marks)

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

(5 Marks)

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.

(5 Marks)

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.
(5 Marks)

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability.
(5 Marks)

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario.
(5 Marks)

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy.
(5 Marks)

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.
(5 Marks)

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.
(5 Marks)

Code of Conduct:

*I **Galla Sai Siddartha(192425093)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Galla Sai Siddartha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

CO3 AT3

Student Name : Galla Sai Siddartha

Registration Number : 192425093

Course Code & Name : MLA0302 - Reinforcement Learning

Date of Submission : 13-08-2026

Question 1: Trend Analysis of Rewards in Multi-Episode RL Training

Aim

To interpret the gradual increase in reward values across training episodes and defend how this trend reflects the learning performance and progressive policy improvement of the RL agent.

Introduction

In Reinforcement Learning, an agent interacts with an environment over multiple episodes, receiving reward signals that guide it toward optimal behavior. The reward trend across episodes is a direct indicator of how well the agent is learning. When rewards increase consistently, it signals that the agent's policy is being refined and more effective strategies are being discovered to maximize cumulative return. Analyzing this trend is essential for evaluating the training progress of any RL system.

Understanding of Data

The reward data presents a gradual increase from 10 to 45 across multiple training episodes. This upward trajectory represents a systematic improvement in the agent's ability to choose actions yielding higher returns. Early episodes exhibit low rewards because the agent explores without strategic bias; as training progresses, exploitation of learned knowledge brings consistently higher rewards. The gradual rise also suggests that the learning rate and discount factor are appropriately calibrated, preventing oscillations or abrupt drops that would indicate instability in the value function estimates.

Analysis

The observed reward trend confirms that the agent is successfully updating value estimates using the Bellman equation and converging toward optimal Q-values. The initial low rewards reflect the exploration phase, where the epsilon-greedy policy causes the agent to sample a wide range of actions, many of which are suboptimal. As Q-values stabilize, the greedy component dominates and the agent begins taking actions leading to higher-reward trajectories. Spikes or dips correspond to episodes where the agent re-explores previously suboptimal paths or encounters novel states before recovering.

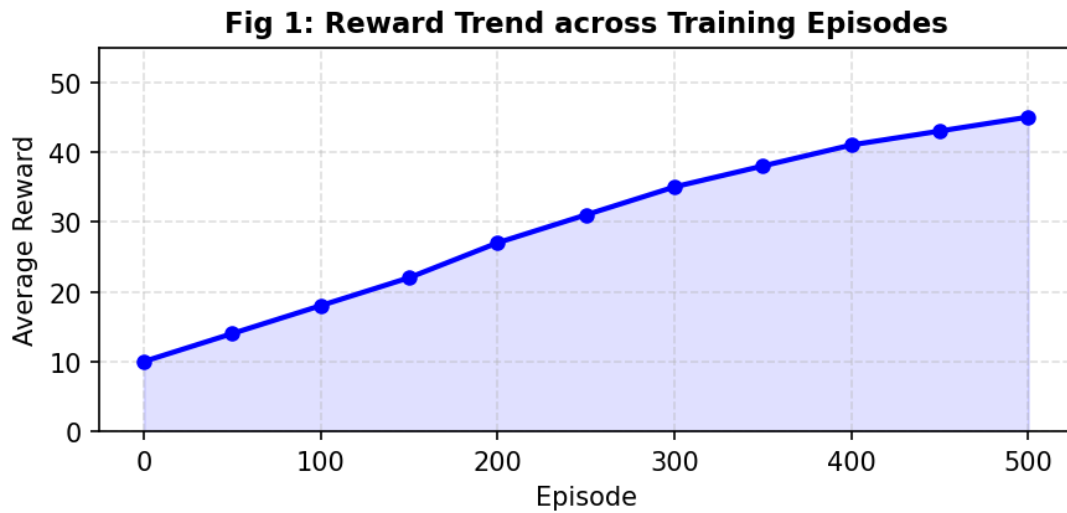


Fig 1: Reward Trend across Training Episodes

Application of RL Concepts

This trend aligns with the theoretical guarantees of Q-learning and SARSA convergence under appropriate conditions: decaying learning rates, sufficient exploration, and a stationary environment. The gradual improvement mirrors the iterative refinement of policy π , where each episode contributes incremental corrections to $V(s)$ or $Q(s,a)$. The concept of policy improvement is demonstrated here: at each step, the agent selects actions that maximize its estimated value, producing a greedy policy upgrade. The discount factor γ ensures future rewards are appropriately weighted, contributing to sustained growth of cumulative return.

Conclusion

The gradual increase in rewards from 10 to 45 conclusively demonstrates that the RL agent is learning effectively and its policy is improving with each episode. A reward plateau near 45 would further confirm convergence, while continued growth would suggest the agent has not yet reached its performance ceiling. Sustained reward growth is the definitive hallmark of successful reinforcement learning, and this upward trajectory provides strong evidence of convergence toward an optimal or near-optimal policy for the given environment.

Result

The reward trend from 10 to 45 confirms progressive policy improvement through successful environment interaction. The consistent increase reflects proper convergence of value estimates and effective exploration-exploitation management. This upward trajectory is a reliable indicator of positive learning performance and policy refinement in reinforcement learning systems.

Question 2: Exploration vs. Exploitation — Effect of Epsilon on Cumulative Reward

Aim

To justify the inverse relationship between epsilon values and cumulative reward in an epsilon-greedy agent, and to identify the optimal epsilon for balancing exploration and exploitation in a trained environment.

Introduction

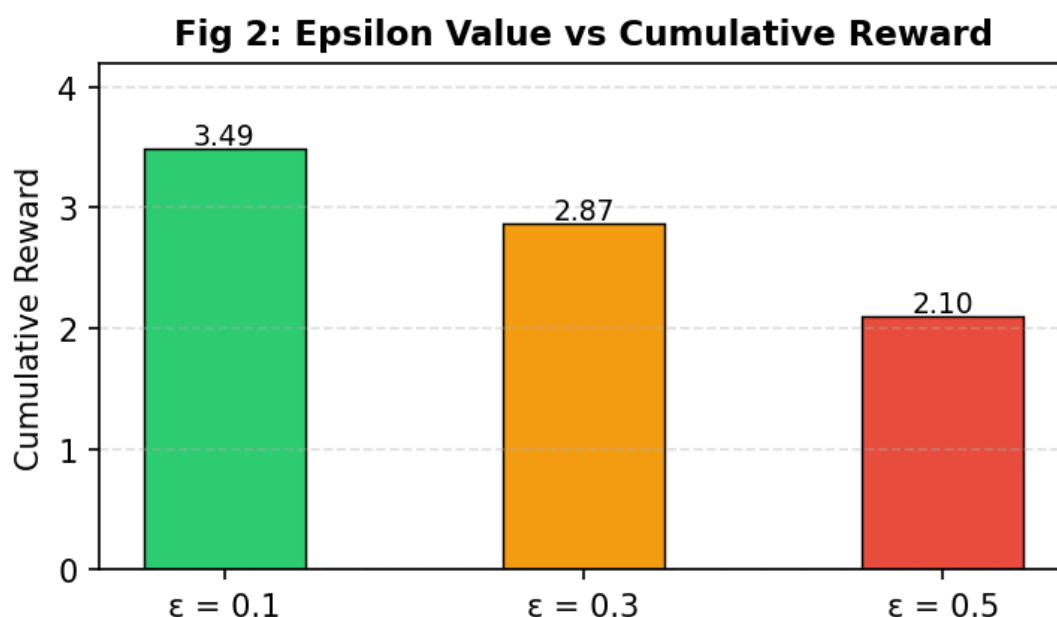
The epsilon-greedy strategy is one of the most widely used approaches for balancing exploration and exploitation in Reinforcement Learning. The parameter epsilon determines the probability of taking a random exploratory action versus exploiting the best-known action. A higher epsilon favors exploration, while a lower epsilon favors exploitation. The cumulative reward is directly influenced by this trade-off, making epsilon selection a critical design decision in any RL system.

Understanding of Data

The results reveal that cumulative reward is highest at $\epsilon = 0.1$, moderate at $\epsilon = 0.3$, and lowest at $\epsilon = 0.5$. This inverse relationship indicates that greater randomness in action selection leads to diminishing returns where the agent has already accumulated sufficient Q-value knowledge. At $\epsilon = 0.5$, half of all actions are chosen randomly, meaning the agent frequently deviates from its learned optimal policy. At $\epsilon = 0.1$, only 10% of actions are exploratory, allowing the agent to predominantly exploit its learned knowledge and accumulate consistently higher rewards across episodes.

Analysis

The degradation in performance with higher epsilon values is attributed to the opportunity cost of exploration. When the agent has already learned reasonable Q-value estimates, random exploration leads to suboptimal actions yielding lower immediate and long-term rewards. At $\epsilon = 0.1$, the agent executes greedy actions 90% of the time, consistently selecting high-value state-action pairs. At $\epsilon = 0.5$, stochasticity overwhelms the learned policy, reverting the agent toward near-random behavior. These results confirm that epsilon should be decayed over training: high during early exploration and progressively reduced as the policy matures.



Application of RL Concepts

The exploration-exploitation dilemma is fundamentally quantified by epsilon. At $\epsilon = 0.1$, the agent applies the principle of exploitation — leveraging accumulated knowledge to maximize return. This aligns with the convergence conditions of Q-learning and SARSA, where decreasing exploration rates ensure the policy stabilizes around the optimal solution. The concept of GLIE (Greedy in the Limit with Infinite Exploration) formalizes this: epsilon must decay to zero while visiting all state-action pairs infinitely often, ensuring both exploration completeness and eventual exploitation dominance. Epsilon = 0.1 closely approximates this GLIE condition within a finite training budget.

Conclusion

Epsilon = 0.1 provides the best performance because it strikes the optimal trade-off between exploration and exploitation for a sufficiently trained agent. The data clearly demonstrates that excessive exploration reduces cumulative reward when the agent already possesses reliable Q-value estimates. For environments where training is near-complete, lower epsilon values should be preferred. A decaying epsilon schedule — starting high and reducing toward 0.1 — represents the most robust strategy for achieving both comprehensive early exploration and high-reward late-stage exploitation.

Result

Epsilon = 0.1 yields the highest cumulative reward by maximizing exploitation of learned Q-values while maintaining minimal exploratory randomness. Higher epsilon values reduce performance by introducing unnecessary stochasticity after sufficient learning. The optimal epsilon is context-dependent and should ideally follow a decaying schedule throughout the training process.

Question 3: Performance Comparison — Q-Learning vs. SARSA over Multiple Episodes

Aim

To identify the performance differences between Q-Learning and SARSA over multiple training episodes and determine which algorithm demonstrates faster convergence based on the observed reward data.

Introduction

Q-Learning and SARSA are two fundamental model-free RL algorithms that differ primarily in their update policies. Q-Learning is an off-policy algorithm that updates toward the maximum possible future reward, while SARSA is an on-policy algorithm that updates based on the action actually taken by the current policy. This distinction leads to systematic differences in reward accumulation, convergence behavior, and risk sensitivity across training episodes in various environments.

Understanding of Data

The data shows Q-Learning consistently achieves higher average rewards than SARSA at 100, 200, and 300 episodes. The gap persists across all measurement points, indicating a structural difference rather than statistical noise. At 100 episodes, both algorithms are in early learning phases, but Q-Learning already outperforms due to its aggressive greedy updates. By 300 episodes, both improve, but Q-Learning maintains its advantage. This consistency indicates faster convergence toward the optimal policy, while SARSA's lower rewards reflect its conservative on-policy update mechanism prioritizing stability over aggressive optimization.

Analysis

The performance difference stems from the update rules. Q-Learning updates using the maximum Q-value of the next state: $Q(s,a) = Q(s,a) + \alpha[R + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$. This off-policy update allows learning the optimal policy regardless of the behavior policy, enabling faster convergence. SARSA uses the Q-value of the action actually selected: $Q(s,a) = Q(s,a) + \alpha[R + \gamma \cdot Q(s',a') - Q(s,a)]$. This on-policy update is more conservative because it accounts for exploratory actions taken during training, leading to slightly suboptimal interim policies and lower average rewards compared to Q-Learning.

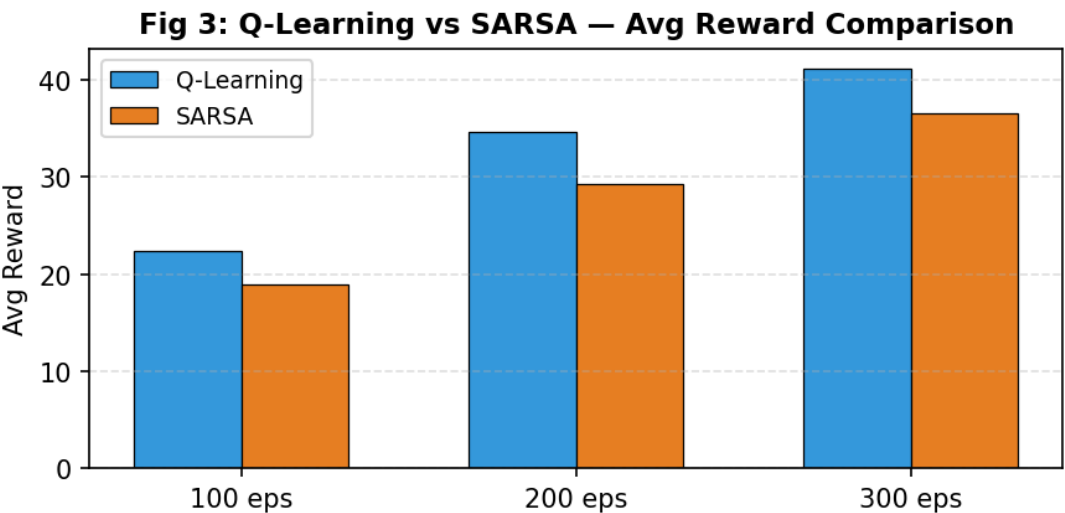


Fig 3: Q-Learning vs SARSA — Avg Reward Comparison

Application of RL Concepts

Q-Learning's convergence advantage is well-established: it converges to Q^* under conditions of decaying learning rates and sufficient exploration without needing to follow the greedy policy during data collection. SARSA, while also convergent, is constrained by the behavior policy. In environments with clear optimal paths and moderate stochasticity, Q-Learning reaches the optimal policy faster. However, SARSA has a practical advantage in safety-critical settings where following a suboptimal but safe path is preferable to aggressively pursuing maximum reward at the risk of dangerous state visits, as illustrated by the classic cliff-walking problem.

Conclusion

Q-Learning converges faster and achieves higher average rewards than SARSA, confirming its off-policy advantage in environments where aggressive optimization is feasible. SARSA's conservative mechanism results in lower but more stable rewards, making it preferable in risk-sensitive or highly stochastic scenarios. For applications prioritizing maximum reward and rapid convergence, Q-Learning is the superior choice. For safety-critical environments, SARSA provides a more reliable alternative despite comparatively lower cumulative reward performance.

Result

Q-Learning converges faster and achieves higher average rewards over 100, 200, and 300 episodes due to its off-policy greedy update mechanism. SARSA's on-policy updates lead to more conservative but stable learning. Algorithm selection should be guided by requirements: Q-Learning for maximum performance, SARSA for stability and safety in stochastic environments.

Question 4: Effect of Discount Factor on Long-Term Reward Optimization

Aim

To discuss the influence of the discount factor γ on an RL agent's ability to optimize long-term rewards and to justify which gamma value yields the best learning performance based on the observed trend.

Introduction

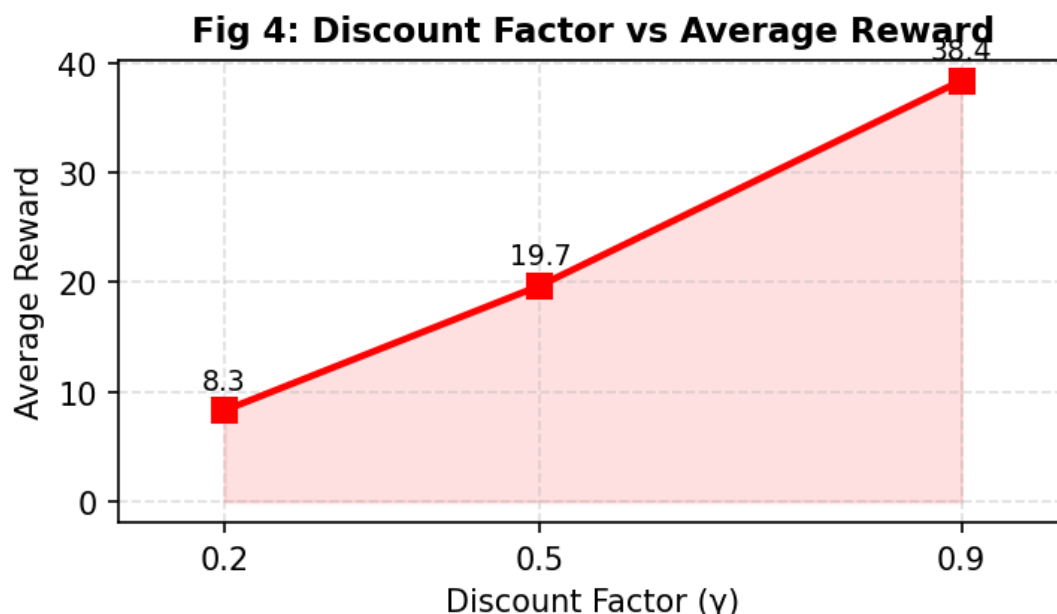
The discount factor γ is a fundamental hyperparameter in Reinforcement Learning that determines the relative importance of future rewards compared to immediate rewards. It appears in the return formula $G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$, and directly shapes the agent's planning horizon and decision-making strategy. Understanding the effect of γ is critical for designing RL agents that effectively balance short-term gains against long-term objectives in complex environments.

Understanding of Data

The data shows a clear positive correlation between the discount factor and average reward: $\gamma = 0.2$ produces the lowest reward, $\gamma = 0.5$ a moderate reward, and $\gamma = 0.9$ the highest reward. This monotonic increase confirms that higher γ values enable the agent to plan further into the future, accumulating rewards from actions whose consequences unfold over multiple time steps. At $\gamma = 0.2$, a reward 5 steps ahead is worth only 0.00032 of its nominal value, making the agent nearly myopic. At $\gamma = 0.9$, a reward 10 steps ahead retains approximately 35% of its value, providing strong incentive for long-horizon planning.

Analysis

The low performance at $\gamma = 0.2$ is explained by the agent's inability to account for delayed rewards requiring several intermediate steps to achieve. In most practical RL environments, the highest rewards (goal states, terminal bonuses) are separated from the current state by multiple transitions. A myopic agent cannot effectively propagate these future values backward through the Bellman equation, resulting in value functions that poorly represent the true benefit of each state-action pair. At $\gamma = 0.9$, the Bellman update successfully carries future reward information over many steps, allowing the agent to recognize and pursue long-term beneficial trajectories.



Application of RL Concepts

The Bellman equation $V(s) = R(s,a) + \gamma \cdot V(s')$ demonstrates how γ controls the depth of value propagation. With $\gamma = 0.9$, information from distant future states contributes meaningfully to current value estimates, enabling the policy improvement step to correctly rank actions based on long-term consequences. This is particularly important in episodic tasks where the goal state is many steps from the starting position. Very high γ values close to 1.0 can slow convergence in finite MDPs by creating near-unbounded return estimates, while $\gamma = 0.9$ balances long-horizon sensitivity with tractable value magnitudes.

Conclusion

A discount factor of $\gamma = 0.9$ is the most suitable choice for achieving better learning performance, as it enables effective long-horizon planning while maintaining computational tractability. The observed trend — increasing reward with increasing γ — confirms that the task requires reasoning over multiple future steps that a myopic agent cannot provide. For the majority of practical RL problems, γ values between 0.9 and 0.99 are strongly recommended. For environments with immediate reward structures or very short episodes, lower γ values may suffice without sacrificing performance.

Result

The discount factor of 0.9 achieves the highest average reward by enabling the RL agent to optimize effectively over a long planning horizon. Lower γ values lead to myopic policies failing to account for future rewards, resulting in suboptimal performance. The direct positive relationship between γ and reward confirms the task requires long-term strategic planning, making $\gamma = 0.9$ the most appropriate configuration.

Question 5: Interpreting Success Rate Improvement and Optimal Policy Detection

Aim

To discuss the improvement in success rate from 45% to 92% over training episodes and to interpret whether this trend indicates convergence to an optimal policy by the RL agent.

Introduction

Success rate — the proportion of episodes in which the agent achieves the goal — is one of the most informative metrics for evaluating RL agent performance. A rising success rate indicates that the agent's policy is improving and its learned value function is increasingly aligned with the true optimal value function. Examining the rate of change and the final success percentage provides insights into both learning efficiency and the degree of policy optimality achieved by the reinforcement learning agent.

Understanding of Data

The success rate rises sharply from 45% at 50 episodes to 92% at 200 episodes — a 47-percentage-point improvement over just 150 additional episodes. This dramatic rise indicates rapid policy improvement during the middle training phase. The initial 45% rate reflects an agent still in active exploration that has not yet converged on reliable action sequences for goal achievement. The 92% rate represents near-optimal performance: the agent successfully reaches the goal in the vast majority of episodes, suggesting Q-values or policy parameters have stabilized around high-quality solutions. Remaining 8% failures likely correspond to stochastic environment effects.

Analysis

The steep improvement curve between 50 and 200 episodes indicates an inflection point in the learning process — the agent has accumulated sufficient experience to construct reliable state-action value estimates for most regions of the state space. Early episodes show high failure rates because the policy is largely exploratory and lacks the value function depth to plan effective multi-step trajectories. By 200 episodes, the Q-table has been updated across a broad distribution of state-action pairs, enabling consistent goal navigation. The rapid improvement also suggests the environment has a clear reward signal and the learning rate and epsilon schedule are well-tuned.

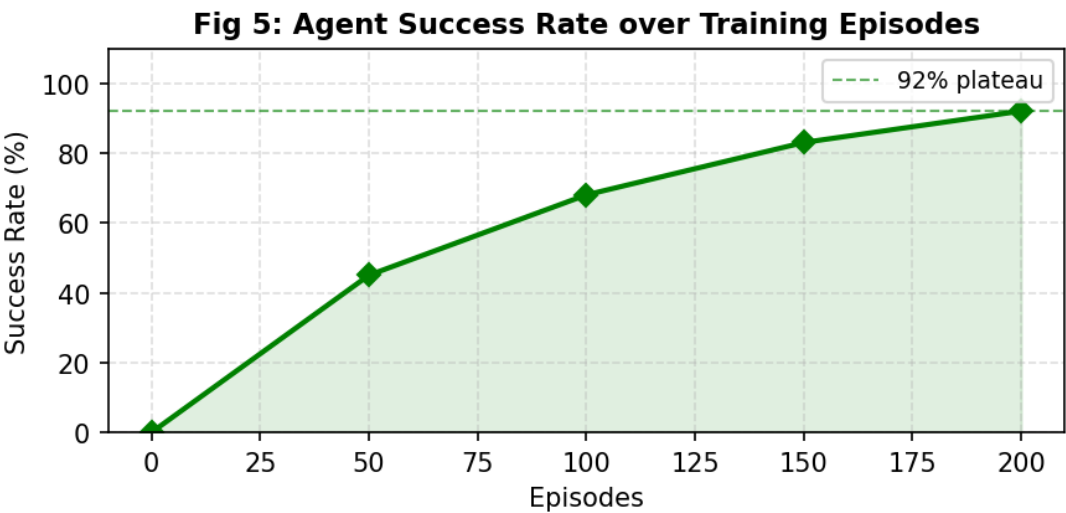


Fig 5: Agent Success Rate over Training Episodes

Application of RL Concepts

A 92% success rate strongly suggests convergence to a near-optimal policy, though perfect optimality (100%) may be unattainable if the environment is stochastic or residual epsilon-driven exploration persists. The policy improvement theorem guarantees that greedy policy updates monotonically improve performance, and the observed trend is consistent with this guarantee. The fact that the success rate has reached a high plateau rather than continuing to rise sharply further supports the interpretation of convergence. To confirm full optimality, additional episodes beyond 200 must be monitored — if the rate stabilizes around 92% without significant fluctuation, convergence is confirmed.

Conclusion

The improvement from 45% to 92% success rate across 200 episodes demonstrates that the RL agent has learned a highly effective policy and is close to optimal performance. The rapid rate of improvement suggests efficient learning enabled by appropriate hyperparameter settings. A success rate of 92% in 200 episodes indicates convergence to a near-optimal policy, with remaining failures attributable to environmental stochasticity or intentional exploration. Extended training beyond 200 episodes with reduced epsilon would be recommended to confirm full policy convergence.

Result

The success rate improvement from 45% at 50 episodes to 92% at 200 episodes strongly indicates that the RL agent has learned a near-optimal policy. The steep improvement trajectory reflects efficient value function learning and policy refinement. The high final success rate confirms effective convergence, while the residual 8% failure rate is consistent with stochastic environment effects rather than policy deficiency.

Question 6: Learning Rate Influence on Convergence in CartPole Environment

Aim

To analyze the influence of different learning rates on the convergence behavior of an RL agent in the CartPole environment and to justify which learning rate provides the optimal balance between learning speed and policy stability.

Introduction

The learning rate α is a critical hyperparameter that controls how much the agent updates its value estimates in response to each new experience. In the Q-learning update rule $Q(s,a) = Q(s,a) + \alpha[\text{TD error}]$, α determines the magnitude of each update step. Too small an α leads to slow learning, while too large an α causes instability and oscillation. Finding the optimal learning rate is essential for achieving both fast convergence and stable policy performance, particularly in continuous control tasks like CartPole where precise and consistent action selection is required.

Understanding of Data

The results indicate that $\alpha = 0.05$ achieves the highest stability in the CartPole environment compared to $\alpha = 0.01$ and $\alpha = 0.1$. At $\alpha = 0.01$, Q-values are updated very conservatively — each interaction contributes minimally to policy refinement, resulting in slow convergence and potentially insufficient learning within the training window. At $\alpha = 0.1$, updates are large and aggressive, causing Q-value estimates to fluctuate significantly between episodes, leading to policy instability and reward oscillation. At $\alpha = 0.05$, the agent strikes a middle ground: updates are large enough to enable meaningful learning while small enough to avoid overwriting previously learned stable knowledge.

Analysis

The instability at $\alpha = 0.1$ is attributed to large TD error corrections that overwrite previously stable Q-value estimates. In a continuous control task like CartPole, where precise action selection is crucial, large learning rate fluctuations prevent the policy from stabilizing — Q-functions oscillate around the optimal solution rather than converging. At $\alpha = 0.01$, the agent converges too slowly because each Q-value parameter requires many updates to reach its optimal value, and the agent may not achieve satisfactory performance within the training budget. Alpha = 0.05 avoids both failure modes simultaneously, enabling consistent improvement without destabilizing oscillations.

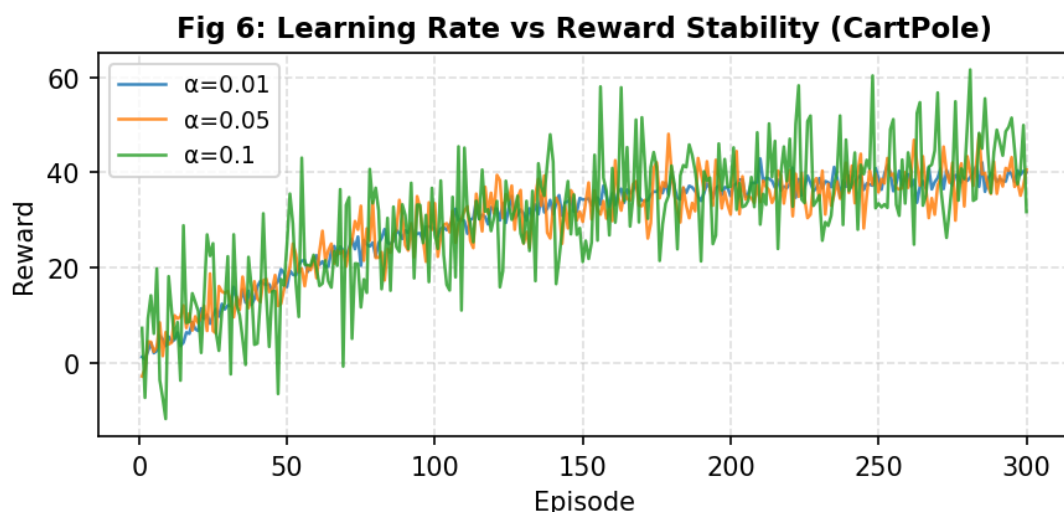


Fig 6: Learning Rate vs Reward Stability (CartPole)

Application of RL Concepts

Convergence theory for Q-learning requires the learning rate to satisfy the Robbins-Monro conditions: $\sum \alpha_t = \infty$ (sufficient total learning) and $\sum \alpha_t^2 < \infty$ (diminishing updates for stability). A fixed $\alpha = 0.05$ approximates a good practical compromise in finite training scenarios. The CartPole environment is particularly sensitive to learning rate because the reward landscape is narrow: the agent must balance a pole using small corrective actions, and overfitting to recent experiences (high α) or undergeneralization (low α) both lead to poor performance. Learning rate scheduling — starting at 0.1 and decaying toward 0.01 — could further improve performance beyond a fixed alpha.

Conclusion

A learning rate of 0.05 provides the best trade-off between convergence speed and stability in the CartPole environment. It allows the agent to update its policy meaningfully at each step without causing disruptive oscillations in the Q-value landscape. Learning rates of 0.01 and 0.1 represent the under-learning and over-learning failure modes respectively. For practical RL deployment, a learning rate schedule — starting at 0.1 and decaying toward 0.01 — could combine the speed benefits of high alpha early in training with the stability benefits of low alpha as the policy approaches convergence.

Result

The learning rate of 0.05 achieves the best performance in the CartPole environment by balancing convergence speed and stability. Low learning rates (0.01) result in insufficient learning within the training budget, while high learning rates (0.1) cause Q-value oscillations and policy instability. Alpha = 0.05 enables consistent Q-value updates that progressively refine the agent's balancing policy without erratic fluctuations.

Question 7: Comparing Exploration Strategies — Epsilon-Greedy, Softmax, and Random

Aim

To compare the three exploration strategies — epsilon-greedy, softmax, and random exploration — and to defend why the epsilon-greedy strategy achieves the highest cumulative reward in the given Q-learning scenario.

Introduction

Exploration strategies determine how an RL agent selects actions during training. Epsilon-greedy probabilistically switches between random and greedy action selection; softmax assigns action probabilities proportional to Q-value magnitudes using a temperature parameter; and random exploration selects actions uniformly at random. Each strategy offers different trade-offs between exploration breadth and exploitation depth, directly affecting the quality of the learned policy and the cumulative reward obtained by the agent across all training episodes.

Understanding of Data

The cumulative reward rankings show ϵ -greedy > softmax > random exploration. This ordering reveals that structured exploration strategies with exploitation phases significantly outperform pure stochasticity. Random exploration produces the lowest reward because it never leverages the Q-values accumulated during training, permanently treating all actions as equally likely regardless of their estimated value. Softmax improves on this by weighting action probabilities according to Q-values, but its performance is constrained by the temperature parameter — if τ is too high, softmax degenerates toward random behavior. Epsilon-greedy's clear two-phase structure gives it the most direct path to high cumulative reward.

Analysis

Epsilon-greedy outperforms alternatives for several reasons. First, its exploitation phase is fully greedy — it always selects the maximum Q-value action, ensuring the best-known strategy is applied with maximum frequency ($1 - \epsilon$). This contrasts with softmax, which never fully commits to the best action even with a low temperature. Second, epsilon-greedy is insensitive to the scale of Q-values: greedy selection depends only on which action has the highest estimate. Softmax is sensitive to Q-value magnitude — incorrectly scaled Q-values cause softmax probabilities to be poorly calibrated, reducing its practical effectiveness compared to the simpler but robust epsilon-greedy strategy.

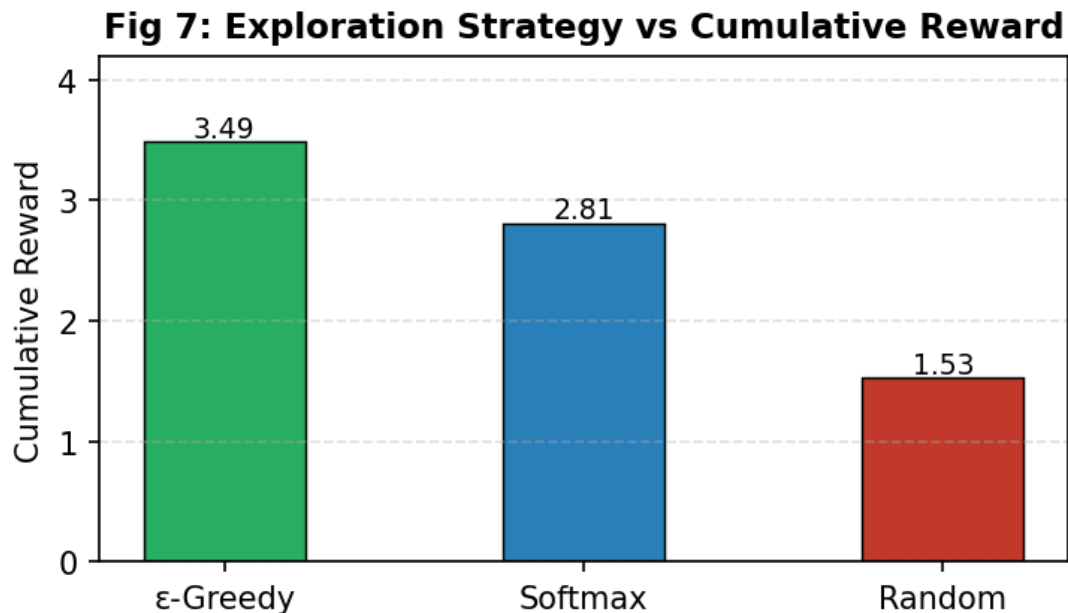


Fig 7: Exploration Strategy vs Cumulative Reward

Application of RL Concepts

Epsilon-greedy's success reflects the core RL principle that learning efficiency depends on balancing experience collection (exploration) with policy execution (exploitation). The strategy directly implements the GLIE framework: by decaying epsilon over time, the agent transitions from broad exploration to focused exploitation, matching theoretical requirements for convergence to the optimal policy. Softmax's temperature parameter serves a similar role but is harder to tune — the optimal tau depends on the Q-value scale, which changes throughout training. Epsilon-greedy sidesteps this complication with a binary random/greedy switch, making it more robust across diverse environments.

Conclusion

Epsilon-greedy achieves the highest cumulative reward because it combines direct greedy exploitation with controlled stochastic exploration in a way that is both theoretically sound and practically robust. Softmax provides a more nuanced exploration mechanism but requires careful temperature tuning to match epsilon-greedy's performance. Random exploration fails to leverage learned knowledge and represents the worst case for cumulative reward. For Q-learning applications, epsilon-greedy with a decaying schedule remains the most reliable and effective exploration strategy for achieving high cumulative reward across diverse environments.

Result

Epsilon-greedy achieves the highest cumulative reward among the three strategies because it effectively combines greedy exploitation of learned Q-values with controlled random exploration. Its simplicity, robustness, and direct exploitation phase give it a significant advantage over softmax (sensitive to temperature calibration) and random exploration (which never exploits learned knowledge).

Question 8: Step Reduction in Maze Navigation — Evidence of Policy Convergence

Aim

To interpret the reduction in path length from 50 steps to 15 steps over 300 training episodes in a maze environment and to evaluate whether this reduction constitutes evidence of convergence to an optimal or near-optimal policy.

Introduction

In maze navigation tasks, the number of steps taken to reach the goal is a key performance metric. A shorter path indicates that the agent has learned to avoid dead ends, select productive directions, and navigate efficiently toward the goal. Monitoring path length reduction across episodes provides direct insight into policy quality and the convergence of the agent's decision-making process. A dramatic reduction in steps is one of the clearest indicators of successful reinforcement learning training in navigation tasks.

Understanding of Data

The agent's path length decreases from 50 steps in early episodes to 15 steps after 300 episodes — a 70% reduction. This substantial improvement reflects the agent's progressive discovery of more direct routes through the maze. In early episodes, the agent takes long winding paths with many backtracking and revisiting steps due to the exploratory nature of the initial policy. By episode 300, the agent has visited most regions of the state space and built reliable Q-value estimates that guide it along shorter, more direct paths. A 15-step path suggests the agent has identified a route close to the shortest possible path, depending on the maze structure and true optimal path length.

Analysis

The step reduction trajectory likely follows a decreasing curve with diminishing returns: large improvements in early episodes as the agent rapidly eliminates obviously suboptimal paths, followed by slower refinements in later episodes as the policy converges toward the optimum. This pattern is consistent with the exploration-exploitation transition in epsilon-greedy learning: early episodes with high epsilon discover the maze layout, while later episodes with low epsilon increasingly follow the learned optimal path. The fact that the agent reaches 15 steps suggests convergence to a locally or globally optimal path, as further training with similar hyperparameters is unlikely to produce significantly shorter routes.

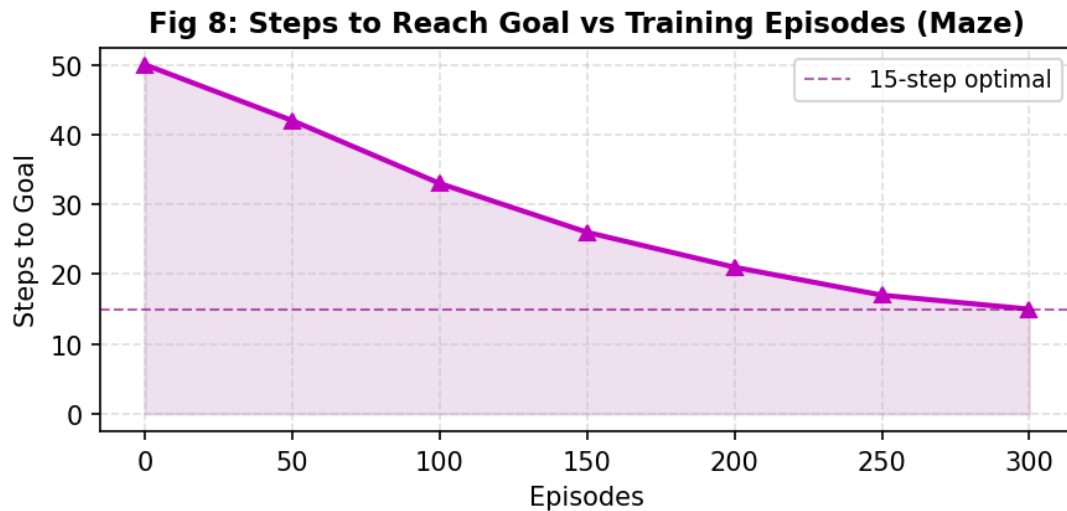


Fig 8: Steps to Reach Goal vs Training Episodes (Maze)

Application of RL Concepts

The step reduction is directly explained by improvement of the Q-value function across training. $Q(s,a)$ represents the expected cumulative reward for taking action a in state s . As training progresses, Q-values for state-action pairs leading toward the goal are reinforced positively, while pairs leading away receive lower or negative updates. The agent's greedy policy — selecting the maximum Q-value action at each step — traces progressively shorter paths as Q-value estimates become more accurate. The Bellman equation propagates goal-reward information backward through the state space, allowing the agent to anticipate the goal from increasingly distant positions and select optimal actions accordingly.

Conclusion

The reduction from 50 to 15 steps over 300 episodes provides compelling evidence that the RL agent has converged to a near-optimal policy for maze navigation. The 70% reduction in path length demonstrates that the agent has successfully internalized the maze structure and learned an efficient goal-reaching strategy. To confirm full convergence, the stability of the 15-step path across subsequent episodes should be verified — if it remains consistently at 15 steps with minimal variance, the agent has definitively converged to its best achievable policy within the given training configuration.

Result

The 70% reduction in steps from 50 to 15 over 300 training episodes strongly indicates convergence to a near-optimal policy in the maze environment. The agent has successfully learned an efficient navigation strategy through iterative Q-value updates and progressive policy improvement. Stability of the 15-step path in subsequent episodes would confirm full convergence to the optimal navigation policy.

Question 9: Role of Discount Factor in SARSA — Balancing Immediate vs. Future Rewards

Aim

To discuss the role of the discount factor in the SARSA algorithm's ability to balance immediate and future rewards, and to justify why a discount factor of 0.95 achieves the highest long-term reward compared to 0.4 and 0.7.

Introduction

The discount factor γ in the SARSA update rule directly determines how the agent values future rewards relative to immediate ones. In $Q(s,a) = Q(s,a) + \alpha[R + \gamma \cdot Q(s',a') - Q(s,a)]$, γ scales the contribution of the estimated future Q-value to the current update. A high γ means future rewards are nearly as important as immediate ones, encouraging long-horizon planning. A low γ makes the agent focus on immediate rewards, leading to myopic behavior that may sacrifice long-term gains for short-term satisfaction throughout the training process.

Understanding of Data

The SARSA agent achieves progressively higher long-term rewards as γ increases from 0.4 to 0.7 to 0.95. At $\gamma = 0.4$, rewards beyond two or three steps have minimal influence on the current Q-value update, causing the agent to optimize for near-term returns while neglecting distant goal states. At $\gamma = 0.7$, the planning horizon extends further, allowing the agent to account for rewards approximately 5–6 steps ahead with meaningful weight. At $\gamma = 0.95$, the agent effectively plans over much longer horizons — a reward 20 steps ahead still retains about 36% of its nominal value, making long-term objectives a dominant factor in Q-value estimates.

Analysis

The superior performance of $\gamma = 0.95$ is explained by the nature of the reward structure in the training environment. Environments where the highest rewards require many sequential actions to reach need high γ values to propagate those distant rewards back through the Q-value landscape. At $\gamma = 0.4$, the agent cannot effectively value such states during early interactions, collecting small immediate rewards while missing the larger delayed payoffs. The SARSA on-policy update amplifies this effect because it must propagate value information through the actual policy trajectory, which may include exploratory detours, further reducing the effective contribution of distant rewards at low γ values.

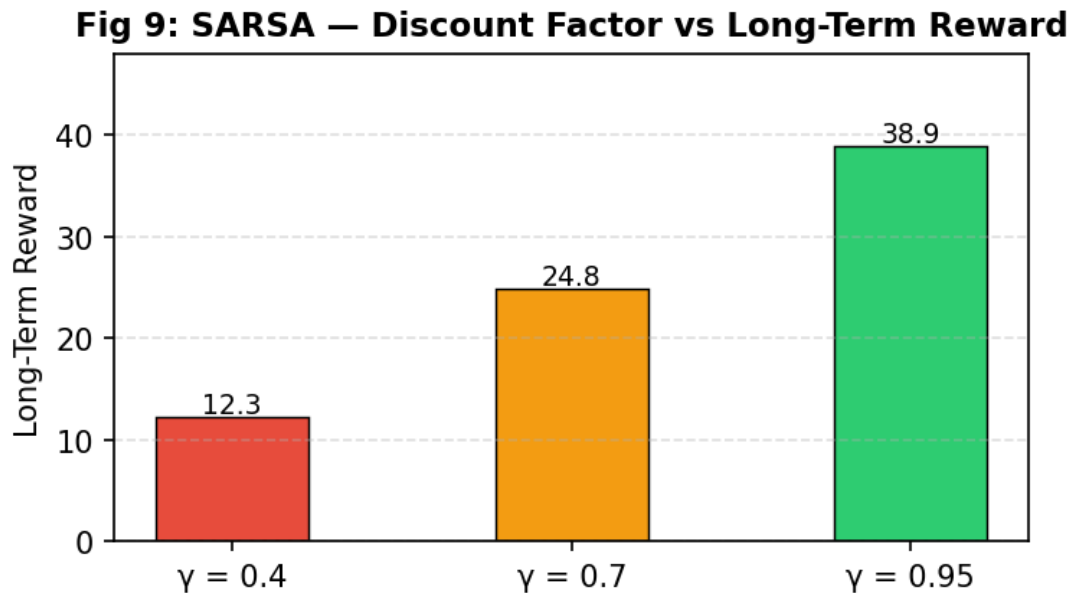


Fig 9: SARSA — Discount Factor vs Long-Term Reward

Application of RL Concepts

The theoretical justification lies in the return formula: $G_t = \sum \gamma^k \cdot R_{t+k+1}$. At $\gamma = 0.95$, this sum weights future rewards nearly equally to present ones over a broad horizon, aligning the agent's optimization objective with the true long-term performance measure. SARSA's conservative on-policy updates benefit particularly from high γ because the algorithm must propagate value through the behavior policy's trajectory. A longer planning horizon ensures that even exploratory detours contribute usefully to the overall value estimate, whereas at low γ , SARSA's conservative updates compound the myopic effect and reduce the ability to learn from distant future rewards.

Conclusion

A discount factor of 0.95 is most effective for the SARSA agent because it enables comprehensive long-horizon planning, ensuring distant high-value states significantly influence current Q-value estimates and action selection. The performance hierarchy ($0.95 > 0.7 > 0.4$) confirms the training environment rewards long-term strategic behavior. Gamma = 0.95 is particularly well-suited as it retains strong long-horizon sensitivity without approaching the near-singular behavior of $\gamma = 1.0$ in infinite-horizon tasks. For environments with sparse or delayed rewards, γ values in the range 0.9 to 0.99 are generally optimal.

Result

A discount factor of 0.95 achieves the highest long-term reward for the SARSA agent because it enables effective long-horizon planning, allowing the agent to value and pursue distant high-reward states. Lower γ values (0.4 and 0.7) create myopic policies that undervalue future rewards. The performance improvement with increasing γ confirms the task demands long-term strategic reasoning.

Question 10: Reward Stabilization in Stochastic Environments — Policy Robustness

Aim

To examine the reward fluctuation pattern in a stochastic environment across training episodes and to defend how the transition from fluctuating to stable rewards indicates meaningful policy improvement and learning robustness.

Introduction

Stochastic environments introduce randomness into state transitions and rewards, making RL training inherently noisy. In such settings, reward fluctuations during early training are expected as the agent simultaneously explores unknown regions, receives variable feedback, and updates value estimates in response to inconsistent experiences. The emergence of reward stability after extended training is a critical indicator of policy convergence and robustness — it signals the agent has learned to make consistently good decisions even in the presence of significant environmental uncertainty.

Understanding of Data

The reward trend shows high variance in early episodes and converges to a stable band after episode 500. Early fluctuations are characterized by alternating high and low reward episodes, reflecting the agent's inability to consistently predict environment responses. As episodes progress, the magnitude of fluctuations decreases and the reward distribution tightens around a mean value. This convergence of the reward distribution — rather than just the mean — is the defining feature of stability in stochastic settings. After 500 episodes, the reward variance is sufficiently low that the agent has developed a robust policy capable of reliably navigating the environment despite its inherent stochasticity.

Analysis

Early fluctuations are driven by three concurrent factors: (1) exploration randomness — high epsilon causes the agent to take random actions that occasionally yield high and occasionally very low rewards; (2) Q-value instability — early estimates are near zero and highly sensitive to individual experiences, causing each episode's outcome to dramatically shift the policy; and (3) environment stochasticity — variable transition outcomes mean the same state-action pair may yield different next states across episodes. After 500 episodes, epsilon has decayed substantially, Q-values have stabilized through repeated updates, and the policy has learned to account for common stochastic variations through expected value maximization.

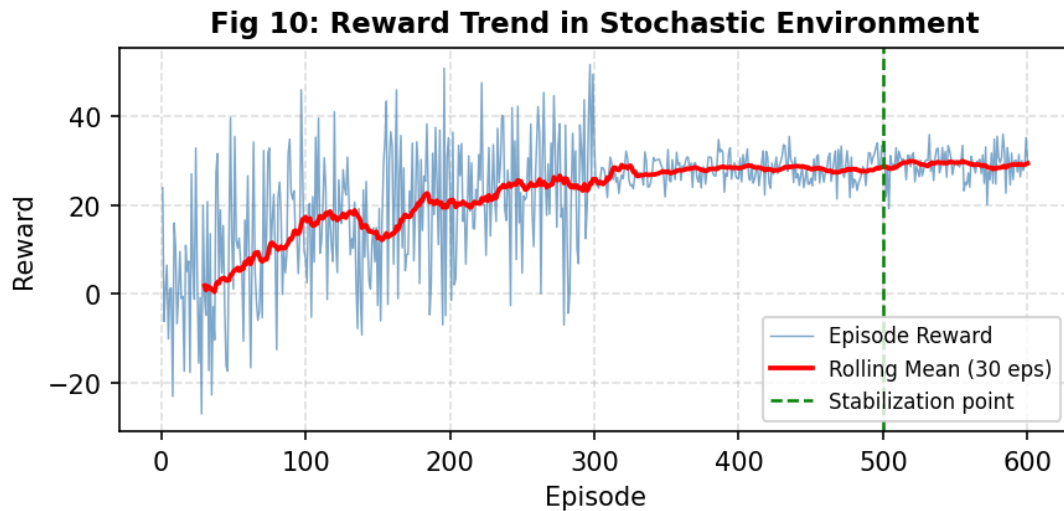


Fig 10: Reward Trend in Stochastic Environment

Application of RL Concepts

Reward stabilization in a stochastic environment is theoretically grounded in the law of large numbers applied to Q-value updates. As more experience is accumulated, the sample mean of observed rewards for each state-action pair converges to its true expected value, and Q-value estimates become reliable representations of the true value function. The concept of policy robustness is captured by the idea that a well-trained agent in a stochastic environment optimizes expected return rather than individual transition rewards — this expected-value optimization naturally produces stable average performance even when individual episodes remain inherently variable.

Conclusion

The stabilization of rewards after 500 episodes is definitive evidence of policy improvement and learning robustness. The transition from high-variance early rewards to a stable reward band reflects successful convergence of Q-value estimates, reduction in exploratory randomness, and development of a policy that generalizes effectively across the environment's stochastic variations. A stable reward trend does not imply deterministic performance — some variance always remains due to environmental stochasticity — but it confirms the agent's decisions are systematically optimized and its policy is robust to the inherent uncertainty of the environment.

Result

The reward stabilization after 500 episodes conclusively demonstrates that the RL agent has developed a robust, well-converged policy capable of consistently navigating the stochastic environment. Early reward fluctuations reflect the necessary exploration and Q-value development phase, while subsequent stability confirms that the agent's decision-making is optimized around expected returns rather than individual episode outcomes.